Muhammad Ali Abbas

Software Engineer — AI Systems — Cloud — Distributed Systems

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Education

NUST College of Electrical and Mechanical Engineering
Bachelor of Computer Engineering *CGPA: 3.47 / 4.00*

English Proficiency

IELTS Academic
Overall Band: 7.5 *Listening: 9.0 — Reading: 6.5 — Writing: 7.0 — Speaking: 7.0*

Professional Experience

Triage AI (Remote) **October 2025 – Present**
Senior Software Engineer *Islamabad, Pakistan*

- Developed a multi-tenant AI-powered healthcare platform supporting pre-admission testing (PAT), automated patient outreach, intelligent clinical workflows, and patient engagement across multiple healthcare organizations.
- Architected distributed workflow orchestration using Python, Temporal, and DynamoDB to manage long-running patient communication, scheduling, clinical triage, and case lifecycle management.
- Built LLM-powered conversational agents using GPT-4, LiteLLM, tool-calling, and safety guardrails to automate two-way patient communication while enabling seamless human escalation for high-risk scenarios.
- Designed asynchronous backend services using Django, Redis, PostgreSQL, and Twilio to support scalable SMS, voice communication, and concurrent patient interactions across isolated tenant environments.
- Integrated MedEngine, Periop, OCR document ingestion, and external healthcare systems to automate document routing, patient data synchronization, and clinical workflow execution.
- Collaborated with cross-functional engineering and clinical teams to design scalable backend architecture, improve reliability, and accelerate delivery of AI-powered healthcare solutions.

Remote (Freelance / Contract) **July 2025 – Present**
Senior Software Engineer *Remote*

- Designed and developed scalable backend services and full-stack web applications for international clients using Python, FastAPI, Django, React, and AWS.
- Implemented distributed workflows using Temporal to orchestrate long-running business processes, background jobs, and asynchronous task execution.
- Developed a cloud-native Patient Management System supporting appointment scheduling, electronic medical records, automated healthcare workflows, and secure authentication.
- Designed highly available cloud infrastructure using AWS services including EC2 and DynamoDB while improving application performance and system reliability.

Labware Arabia **September 2023 – July 2025**
Senior Software Engineer *Dubai, United Arab Emirates*

- Led backend development of a scalable Inspection Management System integrated with LabWare's enterprise platform, enabling automated inspection workflows and regulatory compliance.
- Designed and implemented core backend modules including inspection scheduling, user authentication, role-based access control, compliance tracking, audit trails, and validation workflows using Python and Django.
- Collaborated with product managers and frontend engineers to define system architecture, translate business requirements into technical solutions, and deliver production-ready features.
- Optimized database performance, improved application scalability, conducted code reviews, and integrated third-party enterprise services to enhance operational efficiency.
- Managed deployment, troubleshooting, and production support while ensuring reliable backend performance across multiple customer environments.

Devsinc **May 2022 – August 2023**
Software Engineer *Islamabad, Pakistan*

- Developed full-stack web applications using Python, Django, FastAPI, React, Ruby on Rails, JavaScript, and PostgreSQL for international clients.

- Designed scalable RESTful APIs and responsive frontend applications while following clean architecture and software engineering best practices.
- Built AI-driven social media piracy detection systems capable of identifying infringing content and automatically generating DMCA takedown requests.
- Collaborated within Agile teams, participated in code reviews, and delivered production-quality software with a strong focus on maintainability and performance.

Projects

AI Patient Outreach Platform | *Python, Django, Temporal, GPT-4, LiteLLM, Redis, DynamoDB, PostgreSQL, Twilio*

- Engineered a multi-tenant AI healthcare platform automating patient outreach, appointment reminders, pre/post-operative instructions, follow-up communication, and pre-admission testing workflows across multiple healthcare organizations.
- Architected durable workflow orchestration using Temporal with DynamoDB-backed state management to coordinate long-running patient communication, scheduling, rollout gates, and clinical case lifecycles while ensuring reliability and fault tolerance.
- Developed LLM-powered conversational agents using GPT-4 and LiteLLM with tool-calling, response routing, summarization, safety guardrails, and human escalation while integrating Twilio, MedEngine, Periop, OCR document ingestion, and external healthcare systems.

Inspection Management System | *Python, Django, Django REST Framework, PostgreSQL*

- Developed an enterprise-grade inspection management platform supporting inspection scheduling, compliance tracking, digital checklists, audit trails, and real-time inspection monitoring for large-scale industrial operations.
- Designed secure authentication, role-based access control, workflow validation, and approval processes to ensure regulatory compliance while improving operational transparency and data integrity.
- Optimized backend performance through database tuning, efficient REST API design, and scalable architecture while collaborating closely with frontend engineers and product stakeholders to deliver production-ready features.

Redflag AI | *Ruby on Rails, React, Python, Falcon*

- Developed an AI-powered social media piracy detection platform capable of identifying copyrighted content through automated scanning, keyword analysis, and intelligent content matching pipelines.
- Integrated backend automation using Ruby on Rails to generate, manage, and deliver DMCA takedown notices, significantly reducing manual content enforcement efforts for customers.
- Designed analytics dashboards and reporting modules enabling users to monitor detected infringements, enforcement activity, and platform-wide trends through an intuitive web interface.

Patient Management System | *Python, FastAPI, AWS, DynamoDB, Temporal*

- Developed a scalable healthcare platform supporting patient registration, appointment scheduling, treatment planning, medical record management, and secure clinical workflows.
- Implemented Temporal-based workflow orchestration for appointment reminders, follow-ups, notifications, and other long-running healthcare processes while ensuring reliability and fault tolerance.
- Designed cloud-native infrastructure using AWS services and DynamoDB to deliver highly available, secure, and scalable backend services with authentication, authorization, and audit logging.

Technical Skills

Programming Languages: Python, JavaScript, TypeScript, C++, Ruby, SQL, HTML5, CSS3
Backend Technologies: Django, Django REST Framework, FastAPI, Falcon, Ruby on Rails, Node.js, REST APIs, Microservices, JWT Authentication
Frontend Technologies: React.js, HTML, CSS, Bootstrap, Tailwind CSS
AI & LLM Technologies: GPT-4, LiteLLM, AI Agents, Prompt Engineering, Tool Calling, Retrieval-Augmented Generation (RAG), LLM Integration
Distributed Systems: Temporal, Workflow Orchestration, Asynchronous Programming, Event-Driven Architecture, Multi-Tenant Systems
Databases: PostgreSQL, MySQL, DynamoDB, Redis, MongoDB
Cloud & DevOps: AWS (EC2, DynamoDB, Elastic Beanstalk), Docker, Git, GitHub, CI/CD, Linux
Messaging & Integrations: Twilio, Redis Pub/Sub, MedEngine, Periop, OCR Pipelines, Third-Party API Integration
Software Engineering: Object-Oriented Programming, System Design, Distributed Systems, Design Patterns, Agile/Scrum, Test-Driven Development